

PRACTICAL 1 — Inode, Hard Link and Symbolic Link

Create a folder

```
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical$ mkdir Inode_Practical
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical$ cd Inode_Practical
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ pwd
/home/saireddy/COW_Practical/Redirection_Practical/Inode_Practical
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

Create a file

```
/home/saireddy/COW_Practical/Redirection_Practical/Inode_Practical
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ echo "Hello Linux Inode" > original.txt
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ cat original.txt
Hello Linux Inode
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

Check inode number

```
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ ls -li original.txt
21057 original.txt
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

Use stat

```
21057 original.txt
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ stat original.txt
File: original.txt
Size: 18          Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 21057        Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/saireddy)   Gid: ( 1000/saireddy)
Access: 2026-09-19 05:25:38.548909411 +0000
Modify: 2026-09-19 05:25:27.300909417 +0000
Change: 2026-09-19 05:25:27.300909417 +0000
Birth: 2026-09-19 05:25:27.300909417 +0000
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

Create a HARD LINK

```
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ ln original.txt hardlink.txt
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ ls -li
21057 hardlink.txt 21057 original.txt
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

Check hard-link count

```
21057 hardlink.txt 21057 original.txt
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ stat original.txt
File: original.txt
Size: 18          Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 21057        Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/saireddy)   Gid: ( 1000/saireddy)
Access: 2026-09-19 05:25:38.548909411 +0000
Modify: 2026-09-19 05:25:27.300909417 +0000
Change: 2026-09-19 05:28:19.288898684 +0000
Birth: 2026-09-19 05:25:27.300909417 +0000
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

Create a SYMBOLIC LINK

```
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ ln -s original.txt symlink.txt
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ ls -li
total 8
21057 -rw-r--r-- 2 sairedy sairedy 18 Sep 19 05:25 hardlink.txt
21057 -rw-r--r-- 2 sairedy sairedy 18 Sep 19 05:25 original.txt
21058 lrwxrwxrwx 1 sairedy sairedy 12 Sep 19 05:29 symlink.txt -> original.txt
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

Use find

```
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ find . -inum 123456
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ ./original.txt
./hardlink.txt
-bash: ./original.txt: Permission denied
-bash: ./hardlink.txt: Permission denied
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

Test the links

```
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ echo "New line" >> original.txt
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ cat hardlink.txt
Hello Linux Inode
New line
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

```
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ cat symlink.txt
Hello Linux Inode
New line
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ |
```

PRACTICAL 2 — mmap() File Reading/Writing

Create input file

```
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/Inode_Practical$ cd ..
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical$ mkdir MMap_Practical
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical$ cd MMap_Practical
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/MMap_Practical$ echo "Hello from mmap system call." > data.txt
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/MMap_Practical$ cat data.txt
Hello from mmap system call.
sairedy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/MMap_Practical$ |
```

Create C program

```
GNU nano 7.2 mmap_file.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/stat.h>
#include <sys/types.h>
#include <string.h>

int main()
{
    int fd;
    struct stat st;
    char *data;

    fd = open("data.txt", O_RDONLY);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    fstat(fd, &st);

    data = mmap(NULL, st.st_size,
                PROT_READ | PROT_WRITE,
                MAP_SHARED, fd, 0);

    if (data == MAP_FAILED)
    {
        perror("mmap");
        close(fd);
        return 1;
    }

    printf("Original file content:\n");
    printf("%s\n", (<st.st_size, data);

    memcpy(data, "Memory Mapped I/O!", 19);

    msync(data, st.st_size, MS_SYNC);

    printf("\nFile content after modification:\n");
    printf("%s\n", (<st.st_size, data);

    munmap(data, st.st_size);
    close(fd);

    return 0;
}
```

```
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/MMap_Practical$  
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/MMap_Practical$ gcc mmap_file.c -o mmap_file  
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/MMap_Practical$ ./mmap_file  
Original file content:  
Hello from mmap system call.  
  
File content after modification:  
Memory Mapped I/O!  
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/MMap_Practical$ |
```

Check the actual file

```
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/MMap_Practical$  
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/MMap_Practical$ gcc mmap_file.c -o mmap_file  
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/MMap_Practical$ ./mmap_file  
Original file content:  
Hello from mmap system call.  
  
File content after modification:  
Memory Mapped I/O!  
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/MMap_Practical$ cat data.txt  
Memory Mapped I/O!tem call.  
saireddy@DESKTOP-HKQRL9G:~/COW_Practical/Redirection_Practical/MMap_Practical$ |
```